"The institute, in print"

The imsi exchange

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A Word from the Editor: The Launch of the IMSI EXCHANGE

It is with great enthusiasm that we present the inaugural issue of the *IMSI EXCHANGE*, a publication conceived to serve as a living chronicle of our institute's scientific life. Born from a desire to foster communication across disciplines and to celebrate the diverse voices within our community, this newspaper seeks to bridge the spaces between laboratories, lecture halls, and informal gatherings.

In these pages, you will find research highlights that illuminate the work happening behind laboratory doors, personal reflections that capture the human dimensions of scientific pursuit, and announcements that keep us connected to the vibrant events shaping our academic calendar. The *Exchange* belongs to every member of this institute-teachers who guide, students who discover, staff who sustain, and clubs who create. We invite you to write for it, read it, and share it.

May this modest publication serve as a record of our collective curiosity and a testament to the intellectual spirit that defines IMSI.

Upcoming events

National Conference on Maintenance 4.0: PdM-PHM'26

LGPMI, in collaboration with IMSI and University of Oran 2, invites researchers, academics, and industry professionals to discuss advances in Predictive Maintenance and Prognostics & Health Management within Industry 4.0.

Dates: May 13–14, 2026
Location: Oran, Algeria

Important Dates

Paper Submission: **April 15, 2026**
Acceptance Notification: **April 28, 2026**
Pre-registration: **May 1, 2026**

Topics

- Data and Signal Processing
- Fault Detection and Diagnostics
- Prognostics (RUL)
- Decision Support for Maintenance

Registration: Free for All.

Submit:
`https://pdmphm26.sciencesconf.org/`

Contact:
`lgpmi.conferences@gmail.com`

Institute Seminar

The institute will host a Workday session titled **Matlab: Some Basic Tips & Tricks** on **15 April 2026**. The session will be delivered by Dr. *A. MOULAI-KHATIR*. Attendance is open to all.

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Beyond Grades: Preparing Students for an AI World

In an AI-driven world, knowledge alone is no longer enough; students need creativity, initiative, and social intelligence to truly succeed.

In today's era of Artificial Intelligence, the way we evaluate students needs a fundamental rethink. Traditional exams measure knowledge, but they often fail to capture the skills that will truly define success in the 21st century. Universities should not only focus on academic performance but also foster competencies such as creativity, initiative, organization, social intelligence, and teamwork.

One effective way to achieve this is by actively involving students in cultural, artistic, and sports activities. Participation in these experiences develops leadership, collaboration, and problem-solving skills-qualities that cannot be fully assessed through written tests alone. By incorporating these activities into the evaluation system, universities can encourage holistic development and better prepare students for the complexities of modern life. Education in AI times should therefore prioritize both knowledge and life skills, shaping students who are not only informed but also innovative, socially aware, and capable of taking initiative.

During my time studying in China, I observed an evaluation system that went far beyond grades. While academic awards were given for high marks, the highest honor-"Excellent Student"-was reserved for those who demonstrated well-rounded engagement in university life. To qualify, students needed certificates of participation in volunteer work, workshops, sports teams, musical activities, or cultural events. Recognition was also given for winning trophies or being active members of organizations responsible for ceremonies and campus events.

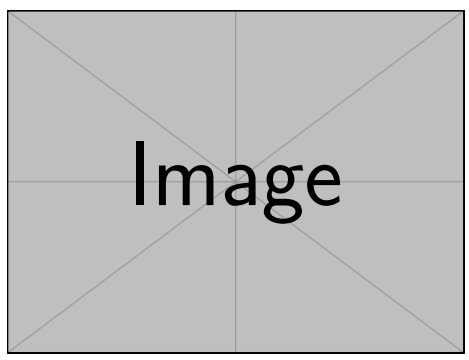
In my first year, I earned high grades and proudly received an academic award. I thought I was on top of the world... until I asked my Egyptian friend why I wasn't considered for the top honor. He laughed and said, "Zahia, you're not in the dance team, not in the committee, not in sports... basically, you're just too good at studying!"

It was an eye-opening moment. The highest awards weren't just about grades; they went to students who balanced academics with cultural, artistic, and sports activities, volunteered, played instruments, and even helped organize ceremonies. That system taught me an important lesson: universities should value initiative, creativity, teamwork, and social engagement-skills that exams alone cannot measure.

After my first-year experience, I decided to get more involved. I joined the African Students Committee, and, truth be told, I was strongly encouraged to participate-my African peers wanted someone of a different color to avoid any appearance of favoritism. I accepted the challenge and took the secretary role, which I shared with Liza from Cameroon.

In the first days, Liza did most of the work while I was learning the ropes. But soon I became effective, helping organize sports events, meetings, and even food competitions. I also helped coordinate and animate workshops-especially those on statistical tools, which were in high demand across many fields. That experience taught me how much students can grow when given responsibility, creativity, and opportunities to take initiative-skills no exam could measure.

Below are some snapshots from my experiences in extra-academic activities. They capture moments that taught me skills beyond the classroom-teamwork, creativity, leadership, and initiative. From organizing sports events and food competitions to running workshops and coordinating cultural activities, these experiences shaped my university journey in ways no exam ever could.



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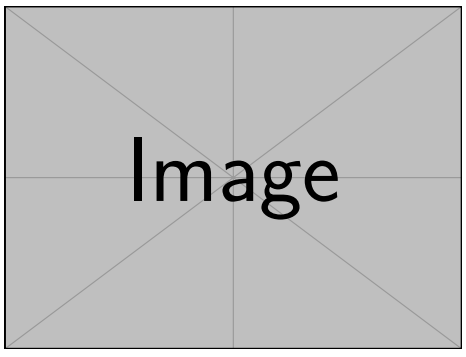

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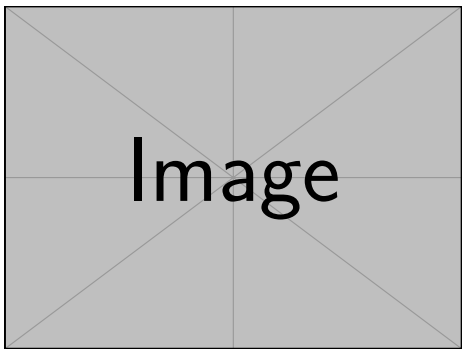

Figure 3: title

I. DROUICHE
Second year engineering student

THE CONCEPT OF A "PROOF" IN MATHEMATICS

"What has been affirmed without proof can also be denied without proof."

Euclid of Alexandria

Mathematics began as a way to quantify our world: to measure land, to predict the motions of the planets, and to keep track of commerce. Ancient mathematicians sought to understand these quantities by asking the question *how*. This led them to observe patterns, which in turn gave rise to mathematical formulas. For instance, the formula for a right triangle $c^2 = a^2 + b^2$ was discovered by noticing this very pattern in geometry (we omit details on the meanings of a, b, and c). We often call it the Pythagorean theorem, but why is Pythagoras the one credited for it, when the relationship existed hundreds of years before him? The answer takes us back to the ancient Greeks. Their methods and philosophical beliefs shaped mathematics in a profound way, leading them to see it through a deeply philosophical lens. They were not satisfied with merely asking *how*; for them, the deeper question was *why*. In seeking to answer it, they created the concept of proof in mathematics. This was a transformative moment. It showed that mathematics is not just about performing calculations, but about understanding the reality of the world around us. The Pythagorean theorem is considered one of the first formulas ever demonstrated in this way, a beautiful testament to the human desire not only to observe, but to truly understand.

So what is a proof? A proof is a collection of arguments that shows whether a claim, which can be a formula or a definition, holds true or false. Such a claim is called a theorem. At first, the methods of demonstration were not yet efficient. Asking the question *why* led to an endless recursion where the mathematician could always ask why again, and again, within the proof itself. It was Euclid of Alexandria who provided a lasting solution. In his book *The Elements*, he established the structure of a rigorous proof, demonstrating 465 theorems that covered nearly all the mathematics known at the time, including geometry and number theory. Remarkably, his structure is still used today. Here is how it works: we begin by laying out the definitions of the field we are studying. Next, we set down foundational principles that are accepted as undeniably true and require no proof which we call postulates or axioms. Then, using logic, we build our mathematics by proving theorems based on these axioms. Because we start with concepts that are beyond doubt, the worry of infinite recursion disappears, and we are free to explore with confidence and clarity.

Here is a simple way to understand it. Imagine we have a group of people, and we want to determine whether each person is handsome or not. First, we define what it means to be handsome and what it means to be ugly, for example, by describing a handsome person or an ugly person with certain traits. Next, we set down our axioms. Now suppose there are two people in the group: Garfield and Bob. Garfield is so extraordinarily handsome that his beauty needs no justification, it is a self-evident truth. We therefore take him as an axiom. Bob, on the other hand, is so unmistakably, almost impressively ugly it's a fact that needs no proof; you only need to look at him to know. So Bob, too, becomes an axiom. From these two axioms, we can then determine whether the rest of the group are handsome or not. For instance, if someone resembles Garfield, then he is handsome, why? Because he looks like Garfield, and Garfield is undeniably handsome. If (unfortunately) someone resembles Bob, then he must be ugly. And if someone appears to resemble neither of our two axioms, we would need to introduce them as a new axiom.

Science is elegant.

Quality Assurance Cell
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QUALITY ASSURANCE AT THE UNIVERSITY: BEYOND THE STANDARD

What does it mean for a university to be certified under ISO 9001:2015? A reflection on processes, people, and purpose.

In recent years, universities across the country have increasingly turned to quality management systems to structure their administrative and academic processes. Among these, the ISO 9001:2015 standard has emerged as a reference framework for institutions seeking to formalize their commitment to continuous improvement, stakeholder satisfaction, and operational excellence.

But what does this standard mean in the context of higher education? And how does it shape the daily experience of those who teach, learn, and work within its walls?

A FRAMEWORK, NOT A PRESCRIPTION

ISO 9001:2015 is fundamentally a process-based standard. It does not dictate what a university should teach or how research should be conducted. Rather, it provides a structure for defining, documenting, and improving the processes that support these core activities. From student admissions to thesis defense, from laboratory procurement to faculty evaluation, the standard encourages institutions to ask a deceptively simple question: *What do we do, and how do we know it works?*

The shift from implicit practice to explicit process can be transformative. Processes that once existed only in the memory of long-serving staff become documented, transferable, and open to scrutiny. Responsibilities that were assumed become assigned. Feedback that was occasional becomes systematic.

THE FOUR PILLARS OF ISO 9001:2015

- **Context of the Organization:** The university must understand its environment-the needs of students, the expectations of employers, the regulatory landscape, and the strategic direction set by its leadership.

- **Leadership:** Quality is not delegated. Top management is expected to demonstrate commitment, establish policy, and create conditions where quality can flourish.

- **Risk-Based Thinking:** Rather than reacting to problems after they occur, the standard encourages anticipation. What might go wrong? How can we prevent it? What opportunities exist that we might otherwise miss?

- **Continuous Improvement:** Certification is not a destination. The standard requires ongoing evaluation of performance, analysis of data, and iterative refinement of processes.

VOICES FROM THE GROUND

For administrative staff, ISO implementation often means clarity. Roles are defined, procedures are documented, and expectations are transparent. For faculty, the impact can be more subtle-streamlined procurement processes, clearer pathways for curriculum modification, and more systematic feedback on teaching and research outputs.

Yet challenges remain. Some faculty express concern that quality management introduces bureaucracy that distracts from core academic missions. Others worry that standardization may stifle the creativity and flexibility that characterize successful research environments. These tensions are real and deserve attention. The most successful implementations treat ISO 9001:2015 not as a constraint but as a scaffold-a structure that supports without constraining.

LOOKING FORWARD

As our institution continues its journey toward certification and beyond, the conversation around quality assurance will evolve. The standard provides a common language and a shared framework, but the substance of quality-the excellence of our teaching, the significance of our research, the satisfaction of our students-remains ours to define and deliver.

The IMSI Exchange invites contributions from all community members on this topic. Have you participated in quality initiatives? What has been your experience with ISO processes? What improvements would you like to see? Your perspectives will help shape how our institution understands and practices quality in the years to come.

Call for Contributions

The IMSI Exchange welcomes contributions from all members of our community-teachers, students, staff, and scientific clubs. Share your voice through any of the following:

- **Research Highlights:** Summarize a recent publication, ongoing project, or significant finding from your laboratory.
  - Tell us what question you asked, how you approached it, and why it matters to your field.
- **Student Reflections:** Share your personal experience navigating research or academic life.
  - Write about your research journey, conference experiences, lessons learned, or advice for fellow students.
- **Club Updates:** Keep the community informed about your scientific club's activities.
  - Report on projects, competitions, outreach efforts, or upcoming meetings. Include calls for new members.
- **Faculty Perspectives:** Offer insights drawn from years of teaching, mentoring, or research.
  - Share reflections on your discipline, guidance for early-career researchers, or thoughts on the future of your field.
- **Staff Voices:** Highlight the essential work that supports research and teaching at IMSI.
  - Describe a day in your role, share practical knowledge, or offer tips for navigating institute resources.
- **Creative Works:** Submit artistic expressions inspired by science.
  - Poetry, short fiction, photography, illustrations, or microscopy images with artistic dimension.
- **Opinion Pieces:** Share your perspective on matters that affect our community.
  - Write about scientific policy, research ethics, education, academic culture, or topics you care about.
- **Announcements:** Spread the word about events and opportunities.
  - Seminars, workshops, conferences, thesis defenses, grant deadlines, or community gatherings.
- **Exchange Corner:** Post classifieds for the community.
  - Textbooks, lab equipment, housing, rideshares, or anything else colleagues might need.
- **Lab Snapshots:** Offer a glimpse into the daily life of a research group.
  - Share photographs with captions, describe a typical day, or introduce the people behind the science.
- **Conference Reports:** Share insights from scientific meetings you attended.
  - Highlight key talks, emerging trends, connections made, or lessons brought back to your work.
- **New Faces:** Welcome recent arrivals to our community.
  - Introduce new faculty members, staff, or students with a brief biography and research interests.
- **How-To Guides:** Share practical knowledge that benefits others.
  - Tips for using equipment, mastering software, writing grants, or navigating administrative processes.
- **Science Communication:** Explain a complex concept for a broad audience.
  - Demystify a theory, clarify a technique, or explore a scientific idea in accessible language.
- **Myth Busters:** Correct common misconceptions in your field.
  - Address misunderstandings that persist in popular culture or even within the scientific community.
- **Interviews:** Share conversations with interesting members of our community.
  - Profile a researcher, staff member, or visiting scholar whose work deserves attention.
- **Photo Essays:** Tell a story through images.
  - Document an experiment, field work, lab life, or community event with a series of photographs and brief captions.
- **Puzzles and Games:** Challenge your colleagues.
  - Submit science crosswords, word searches, riddles, or brain teasers for community enjoyment.
- **Book Reviews:** Recommend readings of scientific interest.
  - Review a recent scientific book, classic text, or even a work of fiction with scientific themes.
- **Behind the Science:** Explore the human side of discovery.
  - Share stories of unexpected findings, persistent challenges, or moments of insight that shaped your work.
- **Career Paths:** Explore the many directions a scientific career can take.
  - Profile alumni working in diverse fields or reflect on your own professional journey.
- **Equipment and Methods:** Share technical innovations.
  - Describe a custom-built setup, a novel protocol, or a clever adaptation of existing tools.
- **Funding Opportunities:** Alert colleagues to grants and fellowships.
  - Highlight deadlines, eligibility, and tips for successful applications.
- **Call for Collaborations:** Seek partners across disciplines.
  - Propose interdisciplinary projects, invite expertise, or suggest shared resources.
- **Science Humor:** Lighten the academic atmosphere.
  - Cartoons, amusing anecdotes from the lab, or playful takes on research life.
- **Questions and Curiosities:** Pose questions that intrigue you.
  - Invite the community to ponder open problems, share speculations, or suggest experiments.
- **Wellness and Balance:** Share thoughts on work-life integration, mental health, or maintaining creativity in demanding environments.

All voices are welcome. Not sure where your idea fits? Make the exchange.